

A Short History of Spacecraft Landings on the Moon

Many years have passed since the first spacecraft landed on the moon in 1966. From then until now, there have been many additional attempts with some succeeding and others failing. Here is a short history of some of these attempts.

First victories

The Soviet Union's Luna 9 spacecraft was the first to land on the moon. It touched down on the lunar surface in 1966. The landing came after several other Soviet spacecraft either did not reach the moon or crashed on the surface.

The American space agency NASA says Luna 9 was the first spacecraft to carry out a soft landing on the moon. It was also the first to send "photographic data" from the moon's surface to Earth. Luna 9's landing proved "the lunar surface could support the weight of a lander and that an object would not sink into a loose layer of dust as some models predicted," NASA said.

In May 1966, the United States followed with its Surveyor 1 mission. NASA describes this effort as "the first of a series of seven robotic spacecraft sent to the moon to gather data in preparation for NASA's Apollo missions." The successful soft landing of Surveyor 1 "was one of the great successes of NASA's early lunar and interplanetary program."

Both the U.S. and Soviet Union had other successful robotic landings. Next, the countries aimed to be the first to land humans on the moon.

In this image provided by NASA, astronaut Buzz Aldrin poses for a photograph beside the U.S. flag deployed on the moon during the Apollo 11 mission on July 20, 1969. (Neil A. Armstrong/NASA via AP, File)

Apollo program

In 1969, NASA successfully landed Apollo 11 astronauts Neil Armstrong and Buzz Aldrin on the moon. The U.S. Apollo program sent 12 astronauts to explore the moon over six missions. Apollo 17 was the last in 1972. The U.S. is still the only country to land humans on the moon. NASA officials say the agency's Artemis program aims to return astronauts to the lunar surface possibly by the end of 2026.

China expands space efforts

In 2013, China became the third country to successfully land a spacecraft on the moon. China landed an exploring vehicle, or rover named Yutu that year. China followed with the Yutu-2

rover in 2019, this time touching down on the moon's unexplored far side.

In 2020, China successfully returned samples of rock and dirt collected by the rovers. The mission returned about 1.7 kilograms of lunar material collected from the near side of the moon. Then in 2024, another mission returned rock and soil samples from the less explored far side of the moon. China has said it aims to land its astronauts on the moon by 2030.

Russian landing failure

In 2023, Russia tried its first moon landing in nearly 50 years. The Luna 25 spacecraft was attempting to land near the lunar south pole. But it is believed to have crash landed on the moon's surface. The country's last effort before that, Luna 24 in 1976, landed successfully and returned moon rocks to Earth.

This handout screen grab taken and released by the Indian Space Research Organisation (ISRO) on August 25, 2023, shows the Chandrayaan-3 rover as it guides itself from the lunar lander to the surface of the Moon. (Photo by ISRO / AFP)

India succeeds on second attempt

After its first lander crashed into the moon in 2019, India made its second attempt in 2023. The country successfully landed its Chandrayaan-3 spacecraft on the lunar surface. The mission made India the fourth nation to complete a moon landing.

Japan lands sideways

In January 2024, Japan became the fifth country to successfully land a spacecraft on the moon. But the SLIM spacecraft landed upside down. This caused some power and communication problems, but the lander was able to operate for a short time.

I'm Bryan Lynn.

The Associated Press reported this story. Bryan Lynn adapted the report for VOA Learning English with additional information from NASA.

Vocabulary

layer - n. ชั้น, ระดับ, จำนวนของสารที่ปกคลุมพื้นผิว

mission - n. (การเดินทางในอวกาศ) ภารกิจ, การบินของยานอวกาศเพื่อปฏิบัติงานหรือภารกิจบางอย่าง

sample - n. ตัวอย่าง, ปริมาณเล็กน้อยของบางสิ่งบางอย่างที่ให้ข้อมูลเกี่ยวกับสิ่งนั้น

Reading Comprehension Quiz

1. According to the article, when was the first spacecraft landing on the moon mentioned in the historical timeline?

- A) 1966
- B) 1969
- C) 2013
- D) 2023

2. Which country achieved the first successful moon landing, according to the article?

- A) United States
- B) Soviet Union
- C) China
- D) India

3. What significant fact did Luna 9's landing prove about the lunar surface, as stated by NASA?

- A) It was covered in a deep layer of dust.
- B) It was incapable of supporting heavy objects.
- C) It could support the weight of a lander.
- D) It contained abundant water ice.

4. What was the primary purpose of the U.S. Surveyor 1 mission in May 1966?

- A) To land humans on the moon.
- B) To collect samples of lunar rock.
- C) To gather data in preparation for NASA's Apollo missions.
- D) To establish a permanent lunar base.

5. After achieving successful robotic landings, what was the next major goal that both the U.S. and the Soviet Union aimed for?

- A) To establish communication with extraterrestrial life.
- B) To land humans on the moon.
- C) To explore other planets in the solar system.
- D) To build space stations in Earth's orbit.

6. How many astronauts did the U.S. Apollo program send to explore the moon in total?

- A) Six
- B) Ten
- C) Twelve

D) Seventeen

7. What is the primary aim of NASA's Artemis program, as mentioned in the article?

- A) To send robotic probes to Mars.
- B) To return astronauts to the lunar surface.
- C) To launch a new space telescope.
- D) To develop faster spacecraft for deep space travel.

8. In what year did China become the third country to successfully land a spacecraft on the moon?

- A) 2013
- B) 2019
- C) 2020
- D) 2024

9. What was a notable characteristic of China's Yutu-2 rover mission in 2019?

- A) It was the first rover to collect water samples.
- B) It landed on the moon's unexplored far side.
- C) It established a permanent base on the lunar surface.
- D) It returned human samples to Earth.

10. Approximately how much lunar material did China successfully return from the near side of the moon in 2020?

- A) 0.7 kilograms
- B) 1.2 kilograms
- C) 1.7 kilograms
- D) 2.0 kilograms

11. By what year has China stated its aim to land its own astronauts on the moon?

- A) 2026
- B) 2028
- C) 2030
- D) 2035

12. What was the outcome of Russia's Luna 25 spacecraft mission in 2023?

- A) It successfully landed near the lunar south pole.
- B) It returned moon rocks to Earth.

- C) It is believed to have crash landed.
- D) It orbited the moon without landing.

13. Which number nation was India to successfully complete a moon landing with its Chandrayaan-3 spacecraft?

- A) Second
- B) Third
- C) Fourth
- D) Fifth

14. What was the unusual characteristic of Japan's SLIM spacecraft landing in January 2024?

- A) It landed on the moon's far side.
- B) It landed upside down.
- C) It deployed multiple rovers.
- D) It discovered water on the surface.

15. Who adapted the report for VOA Learning English?

- A) Neil A. Armstrong
- B) Buzz Aldrin
- C) Bryan Lynn
- D) The Associated Press